

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected, May 2028

East Lansing, MI

- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-Founder

Oct. 2025 - Present

East Lansing, MI

- Engineered serverless automation pipeline integrating GitHub API and Slack SDK to provision team access and workspace invitations, reducing manual onboarding from 15 minutes to <30 seconds for 120+ members.
- Created QR code generation service with AWS S3 storage, CloudFront CDN distribution, and time-based presigned URLs, achieving <100ms global delivery latency with automatic expiration for security.
- Established open source development workflow with GitHub Actions CI/CD, enforcing 85% code coverage via automated test suites and strict linting standards, maintaining comprehensive documentation and contributor guidelines to enable community contributions.

MindMosaic

Product Manager

Aug. 2025 – Nov. 2025

East Lansing, MI

- Directed team of 6 to build full-stack note-taking app with chapter-and-page model, layered Markdown editing, and visual concept mapping via annotated links.
- Architected containerized microservice deployment using Docker Compose, orchestrating FastAPI backend, PostgreSQL database, and Neo4j across isolated containers; established CI/CD pipelines for automated testing and linting to maintain 94% code coverage.
- Integrated PostgreSQL with Neo4j, adding compound B-tree indexes (`user_id`, `created_at`) to reduce note search latency from 850ms to 440ms and cut database CPU utilization by 30% under 50 concurrent queries.

PROJECTS

Mira — Distributed Intelligence Tool | *FastAPI, Next.js, Docker, WebSockets* May 2025 – Nov. 2025

Multi-client intelligence system, analyzing real-time audio streams to execute context-aware automation

- Implemented thread-safe multi-tenant service registry using LRU eviction and TTL-based resource cleanup to manage lifecycle for 100+ concurrent networks, preventing memory leaks in long-running server instances.
- Built production hybrid search pipeline merging BM25 keyword matching with Sentence-BERT embeddings (384-dim), implementing 5-minute conversation boundary detection to optimize context retrieval accuracy.
- Developed Electron desktop client with optimized IPC architecture to stream real-time audio data to the backend, implementing optimistic UI updates and local state management to handle network latency.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow*

Feb. 2025

Won Interactive Media Track with real-time computer vision game utilizing pose-estimation and gesture-detection

- Designed low-latency gesture pipeline using MediaPipe Pose to map 33-point skeletal landmarks to game actions, achieving 60 FPS with sub-20ms input lag by optimizing the inference loop.
- Created custom vector-based physics engine with AABB collision detection to map 3D real-world movements into 2D game state in real-time.

TECHNICAL SKILLS

Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitHub Actions, PostgreSQL, Neo4j
Backend Frameworks FastAPI, SQLAlchemy, Redis, Alembic, Node.js
Frontend Development React, Next.js, Tailwind CSS, Electron, Vite
Languages Python, C++, Java, JavaScript/TypeScript, SQL, Bash